Dataset and Baseline Evaluation Report for Pashto to English and Hindi Neural Machine Translation

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1 Project Overview

This project focuses on building a Neural Machine Translation system for translating Pashto text into English and Hindi. Pashto is a low-resource language, which means that clean and high-quality parallel datasets are limited compared to high-resource languages such as English and Hindi.

In the current stage, I have completed the baseline implementation, dataset collection, dataset cleaning, train-validation-test split, and baseline evaluation. The system uses a pretrained multilingual translation model and compares three translation modes:

1. Pashto to English translation.
2. Direct Pashto to Hindi translation.
3. Pashto to English to Hindi pivot-based translation.

The main goal of this stage was to create a research-level foundation before moving to fine-tuning.

2 Approach Used

The approach used in this project is based on transfer learning using a pretrained multilingual Neural Machine Translation model. Instead of training a model from scratch, I used a pretrained model and evaluated its baseline performance on cleaned Pashto-English data.

The current pipeline is:

Pashto Input Text
$\downarrow$
Text Tokenization using NLLB Tokenizer
$\downarrow$
Pretrained NLLB Transformer Model
$\downarrow$
Pashto to English Translation
$\downarrow$
Direct Pashto to Hindi Translation
$\downarrow$
Pashto to English to Hindi Pivot Translation

This approach was selected because Pashto-Hindi parallel data is limited. Therefore, English is used as a pivot language to improve the Hindi translation pipeline.

3 Pretrained Model Used

The pretrained model used in the current implementation is:

`facebook/nllb-200-distilled-600M`

NLLB stands for *No Language Left Behind*. It is a multilingual Neural Machine Translation model designed to support translation across many languages, including low-resource languages such as Pashto.

The distilled 600M version was selected because it provides a good balance between translation quality and computational feasibility on a normal system.

4 Model Architecture

The model follows a Transformer-based encoder-decoder sequence-to-sequence architecture.

- The encoder reads and understands the input Pashto sentence.
- The decoder generates the translated sentence in the selected target language.
- A forced target language token is used to control the output language.

The architecture used is:

Input Sentence
$\downarrow$
NLLB Tokenizer
$\downarrow$
Transformer Encoder
$\downarrow$
Transformer Decoder
$\downarrow$
Target Language Output

5 Language Codes Used

The NLLB model requires specific language codes for source and target languages. The language codes used in this project are shown in Table 1.

Table 1: Language codes used in the baseline system

Language	Script	NLLB Code
Pashto	Arabic script	`pbt_Arab`
English	Latin script	`eng_Latn`
Hindi	Devanagari script	`hin_Deva`

6 Implementation Details

The implementation was done on a Windows system using Python. A virtual environment was created and the required NLP libraries were installed.

The main libraries used are:

- **Transformers**: for loading the pretrained NLLB model.
- **PyTorch**: for model inference.
- **SentencePiece**: for tokenization support.
- **Pandas**: for dataset handling.
- **Evaluate and SacreBLEU**: for BLEU and chrF evaluation.
- **BeautifulSoup**: for cleaning HTML noise from text.
- **Scikit-learn**: for train-validation-test split.

7 Generation Parameters Used

The following generation parameters were used during baseline translation:

Table 2: Model generation parameters

Parameter	Value
Model	`facebook/nllb-200-distilled-600M`
Maximum sequence length	256 tokens
Decoding method	Beam search
Number of beams	5
Early stopping	True
Target language control	Forced BOS token
Evaluation batch size	2
Evaluation samples	100
Device used	CPU

The forced beginning-of-sentence token was used to control whether the output should be English or Hindi.

8 Dataset Used

For dataset preparation, I used the WMT20 Pashto-English data. The dataset was converted into CSV format with the following columns:

```
pashto, english, source
```

The raw dataset contained noisy sentence pairs, duplicate entries, wrong alignments, HTML/URL noise, and mismatched translations. Therefore, a cleaning pipeline was implemented before evaluation.

9 Dataset Cleaning Pipeline

The dataset cleaning pipeline included the following steps:

1. Removed empty rows.
2. Removed duplicate sentence pairs.
3. Removed URLs and web links.
4. Removed HTML content.
5. Removed very short sentence pairs.
6. Removed very long sentence pairs.
7. Removed sentence pairs with length ratio greater than 4.0.
8. Kept Pashto rows only if Arabic-script characters were present.
9. Kept English rows only if Latin characters were present.

The cleaning parameters used are shown in Table 3.

Table 3: Dataset cleaning parameters

Cleaning Rule	Parameter Used
Minimum sentence length	3 characters
Maximum sentence length	500 characters
Maximum length ratio	4.0
Pashto script check	Arabic Unicode range
English script check	Latin characters
Duplicate removal	Based on Pashto-English pair
URL removal	Regex-based filtering
HTML cleaning	BeautifulSoup

10 Dataset Cleaning Results

After cleaning, the dataset statistics were obtained as shown in Table 4.

Table 4: Dataset cleaning summary

Stage	Number of Rows
Raw rows	93,498
After removing empty rows	93,498
After duplicate removal	93,418
Duplicates removed	80
Final clean rows	90,978
Removed noisy rows	2,440

The final cleaned dataset contains 90,978 Pashto-English sentence pairs.

11 Train, Validation, and Test Split

The cleaned dataset was split into training, validation, and test sets. The split ratio used was approximately:

80% training, 10% validation, 10% testing

The final split is shown in Table 5.

Table 5: Train-validation-test split

Split	Number of Rows
Training set	72,782
Validation set	9,098
Test set	9,098
Total	90,978

12 Evaluation Metrics

The baseline model was evaluated using the following metrics:

- **BLEU**: Measures word-level overlap between model output and reference translation.
- **chrF**: Measures character-level similarity between model output and reference translation.
- **Average inference time**: Measures the average time taken to translate one sentence.

BLEU and chrF were calculated for Pashto-to-English translation because English reference translations were available in the dataset. Hindi BLEU and chrF were not calculated in this stage because Hindi reference translations were not available.

13 Baseline Evaluation Results

The baseline evaluation was performed on 100 test samples using the pretrained NLLB-200 distilled 600M model. The results are shown in Table 6.

Table 6: Baseline evaluation results on 100 samples

Model	Direction	Samples	BLEU	chrF
NLLB 600M	Pashto-English	100	17.97	36.49
NLLB 600M	Pashto-Hindi Direct	100	–	–
NLLB 600M	Pashto-English-Hindi Pivot	100	–	–

The average inference time is shown in Table 7.

Table 7: Average inference time on CPU

Translation Direction	Average Time per Sentence
Pashto-English	4.46 seconds
Pashto-Hindi Direct	5.11 seconds
Pashto-English-Hindi Pivot	4.76 seconds

14 Result Interpretation

The baseline model achieved a BLEU score of 17.97 and a chrF score of 36.49 for Pashto-to-English translation. This shows that the pretrained NLLB model can generate meaningful translations, but the quality is still moderate and requires improvement.

A BLEU score in the range of 10 to 20 generally indicates that the translation is understandable but contains several errors. A chrF score of 36.49 indicates that the model captures some sentence structure and meaning, but the translation still needs improvement.

Therefore, the current model is a good baseline, but it is not yet a final high-quality translation system.

15 Manual Observation from Predictions

Manual inspection of the generated predictions showed that some model outputs were meaningful, but the dataset also contains noisy or mismatched reference translations.

For example, in some cases, the model prediction was closer to the actual Pashto meaning than the reference English sentence. This indicates that the automatic BLEU score may be lower than the actual translation quality because noisy references can reduce the measured score.

This observation is important because it shows that further dataset cleaning and semantic filtering are required before fine-tuning.

16 Current Status

The work completed till now includes:

1. Baseline Pashto to English and Hindi translation system implemented.
2. NLLB-200 distilled 600M model loaded successfully.
3. Pashto-to-English translation implemented.
4. Direct Pashto-to-Hindi translation implemented.
5. Pashto-to-English-to-Hindi pivot translation implemented.
6. WMT20 Pashto-English dataset downloaded and converted into CSV format.
7. Dataset cleaning pipeline implemented.
8. Cleaned dataset created with 90,978 sentence pairs.
9. Train-validation-test split created.
10. Baseline evaluation performed on 100 test samples.
11. BLEU and chrF scores calculated for Pashto-to-English translation.
12. Hindi direct and Hindi pivot translation outputs generated.
13. Initial error analysis started.

17 Limitations Observed

The current baseline system has the following limitations:

- The model is not fine-tuned yet on cleaned Pashto-English data.
- The WMT20 dataset contains noisy and mismatched reference translations.
- Hindi reference translations are not available, so Hindi BLEU and chrF could not be calculated.
- Inference is slow on CPU.
- Some translations miss words, change meaning, or generate unnatural Hindi.
- Direct Pashto-to-Hindi translation is weaker than the pivot-based method in some cases.

18 Next Step

Before fine-tuning, the next step is to improve the evaluation quality by creating a manually checked test set of 50 to 100 Pashto-English-Hindi sentence pairs. After that, the model can be fine-tuned on cleaned Pashto-English data.

The planned next steps are:

1. Create a manually verified test set.
2. Add semantic similarity filtering using LaBSE, LASER, or multilingual sentence embeddings.
3. Fine-tune NLLB on 10k cleaned Pashto-English sentence pairs.
4. Fine-tune NLLB on 50k cleaned Pashto-English sentence pairs.
5. Compare baseline NLLB with fine-tuned NLLB.
6. Evaluate using BLEU, chrF, inference time, and manual human scoring.
7. Add IndicTrans2 for better English-to-Hindi translation.
8. Compare direct Hindi translation with pivot-based Hindi translation.

19 Conclusion

In this stage, I successfully moved the project from a simple translation demo to a research-level dataset and evaluation foundation. The pretrained NLLB-200 distilled 600M model was used as the baseline system. The WMT20 Pashto-English dataset was collected, cleaned, and split into train, validation, and test sets.

The baseline model achieved a BLEU score of 17.97 and a chrF score of 36.49 on 100 Pashto-English test samples. These results show that the baseline model is able to generate meaningful translations, but the quality is still moderate. The next stage will focus on better dataset filtering, manual evaluation, and fine-tuning the NLLB model to improve translation accuracy.